swivel

the safety companion for female & nonbinary USC students

by: Iriza Mushibuka, Maia Nkonabang, & Halla Ibrahim

meet the team:

Team #6 - Team Swivel



Maia Nkonabang

- PDP in Computer Engineering
- 22 y.o
- fun fact: I had a red devil fish names Benjamin that died of alcohol posioning



Iriza Mushibuka

- B.S Mechanical Engineering
- 22 y.o
- fun fact: I was named after a cow!



Halla Ibrahim

- B.F.A Interactive Media & Game Design
- 22 y.o
- fun fact: I speak 4 languages!

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stats:

In 2019, The Assn. of American Universities reported that female USC students **reported higher numbers than the national average** of college sexual assault cases

@ stanford:

23.8%

of undergrads reported sexual assault during college years

@ usc:

31%

of undergrads reported sexual assault during college years

@ caltech:

14.3%

of undergrads reported sexual assault during college years

aau campus climate survey (2019). association of american universities (aau). (2019, october 15). https://www.aau.edu/key-issues/campus-climate-and-safety/aau-campus-climate-survey-2019

problem:

Female & Non-binary USC Students don't feel safe when walking alone on or near campus

*They must always be on high alert but that can be hard after a long day of **classes, extracurriculars, and work**

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How Might We...

Help women and Non-binary students feel safe when walking alone near the USC Campus

solution:

meet...

swivel

Your campus safety companion!

how it works:

start: A student is walking alone at night

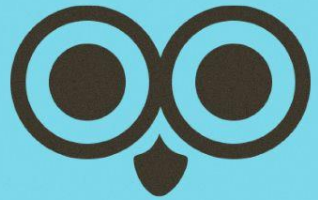
A person walks within 2 feet behind you

swivel's buzzer will go off

if swivel's button isn't pressed after more than 10 seconds

an emergency contact's phone will be sent a message alerting them to check in on you

after 1m 30 sec , a follow-up message with DPS' phone number will be sent



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how we made it:

our components



usb-c battery pack



button



esp 32



sensory distance module



buzzer



bread board

swivel

our process:

//how we built it:

- ultrasonic range sensor to measure distance from an object (person in this case)

- buzzer to alert the user that someone has been within a certain range of them for a certain amount of time

- button to turn off buzzer and notification if false alarm

- timer logic to calculate when someone has been behind you for a certain time + sending messages when the buzzer has went off for 10 seconds

//challenges we ran into:

- coming up with realistic durations of time for when we want a message sent

- making the product durable and transportable

- camera system was buggy

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why swivel:

Why choose swivel?

peace of mind

small, portable & easily accessible!

You don't always have to keep your head on a **swivel**

tool for keeping your loved ones safe

eases anxiety when walking alone, especially at night

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accomplishments & future considerations:

// accomplishments that we're proud of

1. making the product into a backpack sort of shape to transport around
2. coming up with a catchy name and logo for the product
3. creating the timer logic
4. integrating iot with messaging system

// future considerations

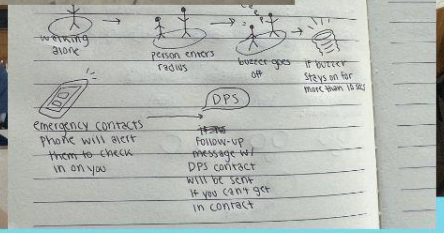
1. integrating a camera to take pictures of the culprit and sending to trusted friends
2. adjusting buzzer volume level (by using an actual sound sensor)
3. fine tuning text messaging system so second text isn't sent if the person isn't in danger
4. better range of danger with levels (like high danger ~2m and medium danger ~5m)

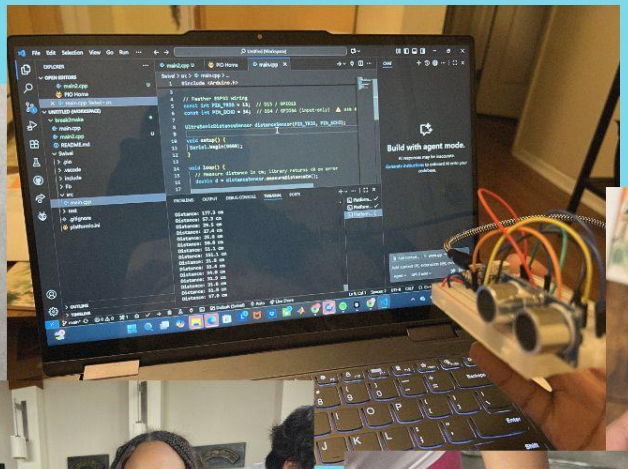
our process in pictures! :



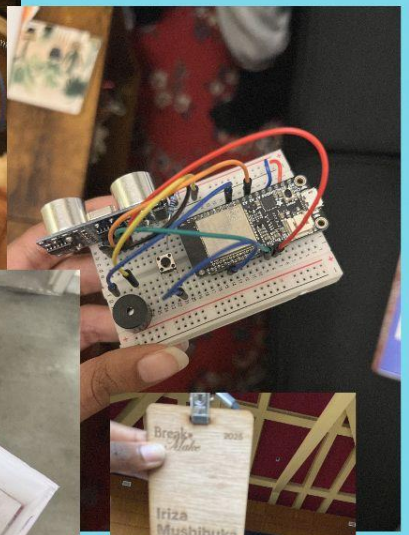
















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